

Tableau Capstone Project: Retail Sales Data

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Stock ID: 47857043



Introduction

Introduction

- This project delivers an end-to-end Tableau workflow for retail sales (2017–2019).
- Six CSV sources (sales, product hierarchy, stores, cities, and name lookups) were integrated, cleaned (dates, nulls, formats, text), and modeled through relationships on product, store, and city keys.
- We built calculated fields (Sales Growth Percentage and Average Stock Levels) and key visuals: time-series performance, regional analysis, sales by product, and sales by store.
- The outcome is a reliable, reusable model that surfaces trends, compares cities and stores, and supports evidence-based commercial decisions.





Methodology

Data Details

The data would normally come from the company's own systems, but in this case we have taken the data from the Retail Sales Data data set at the Kaggle website.

Retail Sales Data

In the Kaggle data set, you will find three CSV files.

- sales.csv - Daily sales data covering 2017-2019.
- product_hierarchy.csv - Data containing the hierarchy and sizes of the products.
- store_cities.csv - Data containing the store's city, type, and size information.

* The dataset obtained from Kaggle contains 19.5 million entries. If we were to use the entire dataset, cleaning and other operations would become prohibitively time-consuming. For this reason, the sales.csv file has been truncated to approximately 600,000 rows of data for your use.

Data Details

- To enhance data understanding, readability, and analysis, we have included three additional CSV files and the existing three CSV files.
- `product_names.csv` - This file links product identifiers with their corresponding names, providing a mapping for easier reference and analysis.
- `store_names.csv` - This file associates store identifiers with their respective names, allowing for better identification and analysis of stores.
- `city_names.csv` - This file maps city identifiers to their names, enabling geographical analysis and insights based on city-level data.

Methodology

- **Strict step-by-step execution.** The labs were followed exactly as written: import, explore, clean, model relationships, build calculations, and create visualizations.
- **Targeted self-learning.** When unsure, open sources (e.g., Google/Tableau help) were consulted to understand the concept and proceed without blocking the workflow.
- **Standardization & traceability.** Every decision was documented with before/after screenshots (nulls, formats, text).
- **Systematic cleaning.** Null detection, targeted imputations (cluster-level means for physical dimensions), text normalization (TRIM/UPPER), and Data Interpreter to remove empty rows/headers.
- **Robust relational model.** Logical relationships on product/store/city keys, cardinality and uniqueness checks (CNTD), and orphan control before enabling referential integrity.
- **Goal-driven visuals.** Line, bar, scatter, and packed-bubble charts with enriched tooltips; color/size configured for quick insight and visual consistency.
- **Final quality gate.** Consistency checks across Revenue, Sales, and Stock; duplicate prevention due to joins; saved .twbx workbook and an evidence checklist for the presentation.

Module 1: Preparing Data and Creating Basic Visualizations

Clean the Sales Data – Import the sales data

Tableau Public - Libro1

Archivo Datos Ventana Ayuda

Tableau Desktop Public Edition

Adquirir Tableau

Conexiones [Añadir](#)

sales

Archivos

☐ Usar el intérprete de datos

Puede que el intérprete de datos esté disponible para limpiar su libro de trabajo Archivo de texto.

- city_names.csv
- product_hierarchy.csv
- product_names.csv
- sales.csv
- store_cities.csv
- store_names.csv

[Nueva unión de filas](#)

[Nueva extensión de tabla](#)

sales

sales.csv

Relacionar tablas

Arrastre tablas cerca sales.csv para establecer relaciones entre ellas

Añadir una tabla base

Arrastre tablas base adicionales para el análisis multifactor.

¿Necesita ayuda para añadir datos? [Más información](#)

sales.csv 13 campos 29999 filas

100 filas

Nombre	sales.csv
Product Id	
Store Id	
Date	
Sales	
Revenue	
Stock	
Price	
Promo Type 1	
Promo Bin 1	
Promo Type 2	
Promo Bin 2	

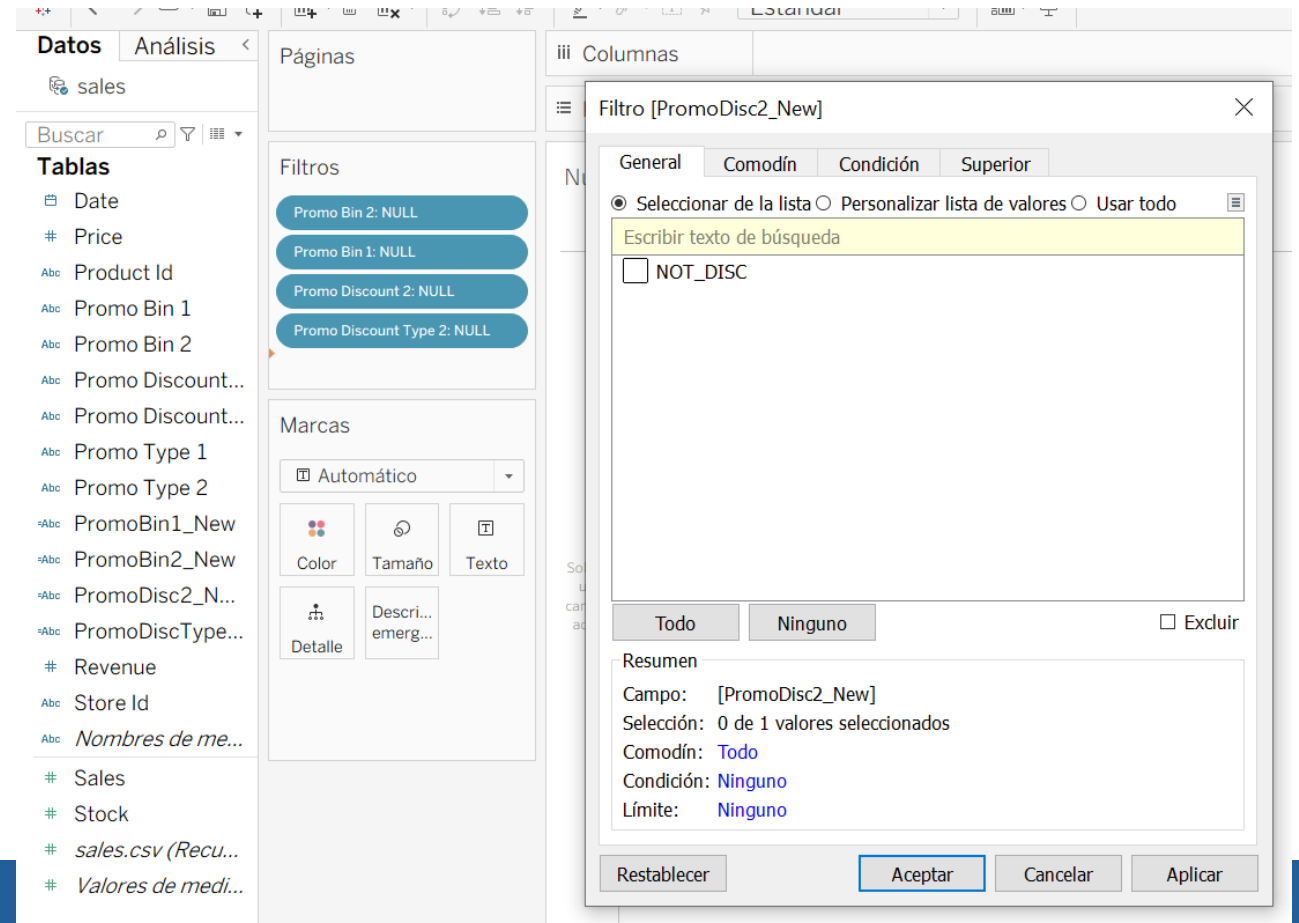
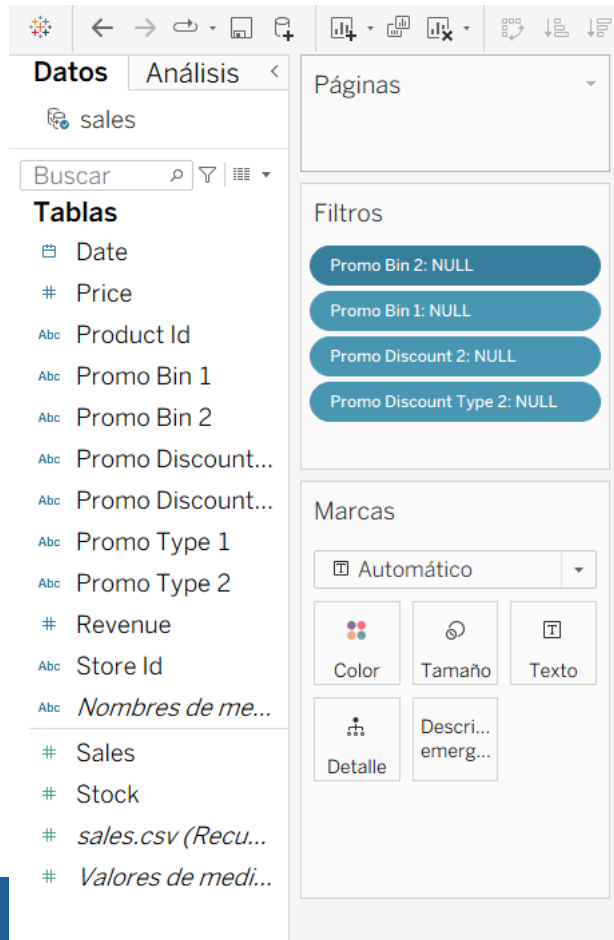
Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	Product Id	sales.csv	product_id
Abc	Store Id	sales.csv	store_id
Cal	Date	sales.csv	date
#	Sales	sales.csv	sales

Abc sales.csv	Abc sales.csv	Cal sales.csv	# sales.csv	Abc sales.csv	# sales.csv	Abc sales.csv	Abc sales.csv	Abc sales.csv	Abc sales.csv	Abc sales.csv
Product Id	Store Id	Date	Sales	Revenue	Stock	Price	Promo Type 1	Promo Bin 1	Promo Type 2	Promo Bin 2
P0005	S0001	02/01/2017	0	0	7	33.9	PR14	NULL	PR03	NULL
P0011	S0001	02/01/2017	0	0	10	49.9	PR14	NULL	PR03	NULL
P0015	S0001	02/01/2017	1	2.41	20	2.6	PR14	NULL	PR03	NULL
P0017	S0001	02/01/2017	0	0	13	1.49	PR14	NULL	PR03	NULL
P0018	S0001	02/01/2017	0	0	49	1.95	PR14	NULL	PR03	NULL
P0024	S0001	02/01/2017	0	0	23	1.95	PR14	NULL	PR03	NULL
P0035	S0001	02/01/2017	2	4.54	14	2.45	PR14	NULL	PR03	NULL
P0046	S0001	02/01/2017	0	0	4	34.5	PR14	NULL	PR03	NULL

Fuente d... Hoja 1

Module 1: Preparing Data and Creating Basic Visualizations

Clean the Sales Data – Identify missing values – BEFORE AND AFTER



Module 1: Preparing Data and Creating Basic Visualizations

Clean the Sales Data –Format the data – BEFORE AND AFTER

The screenshot displays the Tableau Desktop Public Edition interface. On the left, the 'Datos' (Data) pane shows the 'sales' data source. Below it, the 'Tablas' (Tables) pane lists various fields: Date, Price, Product Id, Promo Bin 1, Promo Bin 2, Promo Discount..., Promo Type 1, Promo Type 2, PromoBin1_New, PromoBin2_New, PromoDisc2_N..., PromoDiscType..., Revenue, Store Id, and Nombres de me... (likely 'Nombre de medida'). The 'Filtros' (Filters) pane shows filters for 'Promo Bin 2: NULL', 'Promo Bin 1: NULL', 'Promo Discount 2: NULL', and 'Promo Discount Type 2: NULL'. The 'Marcas' (Marks) pane shows the 'Automático' (Automatic) mark type. The 'Columnas' (Columns) pane shows the 'sales.csv' data source. The 'Filtro [PromoDisc2_N...]' pane shows the 'General' tab with 'Seleccionar de la lista' (Select from list) and 'Escribir texto de búsqueda' (Write search text) options. The 'Conexiones' (Connections) pane shows a list of files: city_names.csv, product_hierarchy.csv, product_names.csv, sales.csv, store_cities.csv, and store_names.csv. The 'Archivos' (Files) pane shows a list of files: city_names.csv, product_hierarchy.csv, product_names.csv, sales.csv, store_cities.csv, and store_names.csv. The 'Nueva unión de filas' (New union of rows) button is visible. The 'Nueva extensión de tabla' (New table extension) button is visible. The 'Resumen' (Summary) pane shows 'Campo: [PromoDisc2_N...]', 'Selección: 0 de 1 v', 'Comodín: Todo', 'Condición: Ninguno', and 'Límite: Ninguno'. The 'Restablecer' (Reset), 'Aceptar' (Accept), 'Cancelar' (Cancel), and 'Aplicar' (Apply) buttons are at the bottom. The main view shows a data table with columns: Product Id, Store Id, Date, Sales, Revenue, Stock, Price, Promo Type 1, Promo Bin 1, PromoBin1_New, and Promo 1. The table contains 17 rows of data.

Product Id	Store Id	Date	Sales	Revenue	Stock	Price	Promo Type 1	Promo Bin 1	PromoBin1_New	Promo 1
P0005	S0001	01/02/2017	0	0.00	7	33.90	PR14	NULL	NOT_BIN	PR03
P0011	S0001	01/02/2017	0	0.00	10	49.90	PR14	NULL	NOT_BIN	PR03
P0015	S0001	01/02/2017	1	2.41	20	2.60	PR14	NULL	NOT_BIN	PR03
P0017	S0001	01/02/2017	0	0.00	13	1.49	PR14	NULL	NOT_BIN	PR03
P0018	S0001	01/02/2017	0	0.00	49	1.95	PR14	NULL	NOT_BIN	PR03
P0024	S0001	01/02/2017	0	0.00	23	1.95	PR14	NULL	NOT_BIN	PR03
P0035	S0001	01/02/2017	2	4.54	14	2.45	PR14	NULL	NOT_BIN	PR03
P0046	S0001	01/02/2017	0	0.00	4	34.50	PR14	NULL	NOT_BIN	PR03

Module 1: Preparing Data and Creating Basic Visualizations

Clean the Sales Data – Standardize text

¿Necesita ayuda para añadir datos? [Más información](#)

sales.csv

17 campos 29999 filas

100 → filas

Nombre

sales.csv

Campos

Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	Product Id	sales.csv	product_id

Abc sales.csv	Abc sales.csv	sales.csv	sales.csv	sales.csv	sales.csv	sales.csv	Abc sales.csv	Abc sales.csv	Abc Cálculo	Abc sales.csv
Product Id	Store Id	Date	Sales	Revenue	Stock	Price	Promo Type 1	Promo Bin 1	PromoBin1_New	Promo 1
P0005	S0001	01/02/2017	0	0.00	7	33.90	PR14	NULL	NOT_BIN	PR03
P0011	S0001	01/02/2017	0	0.00	10	49.90	PR14	NULL	NOT_BIN	PR03
P0015	S0001	01/02/2017	1	2.41	20	2.60	PR14	NULL	NOT_BIN	PR03
P0017	S0001	01/02/2017	0	0.00	13	1.49	PR14	NULL	NOT_BIN	PR03
P0018	S0001	01/02/2017	0	0.00	49	1.95	PR14	NULL	NOT BIN	PR03

¿Necesita ayuda para añadir datos? [Más información](#)

sales.csv

19 campos 29999 filas

100 → filas ⚙️

Nombre

sales.csv

Campos

Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	Product Id	sales.csv	product_id
Abc	Product Id_New	Cálculo	Calculation_1519...
Abc	Store Id	sales.csv	store_id
Abc	Store Id_New	Cálculo	Calculation_1519...

Abc sales.csv	Abc Cálculo	Abc sales.csv	Abc Cálculo	Cal sales.csv	# sales.csv	# sales.csv	# sales.csv	# sales.csv	Abc sales.csv	Abc sales.csv
Product Id	Product Id_New	Store Id	Store Id_New	Date	Sales	Revenue	Stock	Price	Promo Type 1	Promo Bin 1
P0005	P0005	S0001	S0001	01/02/2017	0	0.00	7	33.90	PR14	NULL
P0011	P0011	S0001	S0001	01/02/2017	0	0.00	10	49.90	PR14	NULL
P0015	P0015	S0001	S0001	01/02/2017	1	2.41	20	2.60	PR14	NULL
P0017	P0017	S0001	S0001	01/02/2017	0	0.00	13	1.49	PR14	NULL
P0018	P0018	S0001	S0001	01/02/2017	0	0.00	49	1.95	PR14	NULL
P0024	P0024	S0001	S0001	01/02/2017	0	0.00	23	1.95	PR14	NULL
P0035	P0035	S0001	S0001	01/02/2017	2	4.54	14	2.45	PR14	NULL
P0046	P0046	S0001	S0001	01/02/2017	0	0.00	4	34.50	PR14	NULL

Module 1: Preparing Data and Creating Basic Visualizations

Clean the Product Hierarchy Data . Imported

Archivo Datos Ventana Ayuda

Tableau Desktop Public Edition Adquirir Tableau

product_hierarchy

product_hierarchy.csv

Relacionar tablas
Arrastre tablas cerca product_hierarchy.csv para establecer relaciones entre ellas

Añadir una tabla base
Arrastre tablas base adicionales para el análisis multifactor.

¿Necesita ayuda para añadir datos? [Más información](#)

product_hierarchy.csv 10 campos 699 filas

100 → filas

Nombre	product_hierarchy.csv
product_id	product_id
product_length	product_length
product_depth	product_depth
product_width	product_width

product_id	product_length	product_depth	product_width	cluster_id	hierarchy1_id	hierarchy2_id	hierarchy3_id	hierarchy4_id
P0000	5.0000	20.0000	12.0000	NULL	H00	H0004	H000401	H00040105
P0001	13.5000	22.0000	20.0000	cluster_5	H01	H0105	H010501	H01050100
P0002	22.0000	40.0000	22.0000	cluster_0	H03	H0315	H031508	H03150800
P0004	2.0000	13.0000	4.0000	cluster_3	H03	H0314	H031405	H03140500
P0005	16.0000	30.0000	16.0000	cluster_9	H03	H0312	H031211	H03121109
P0006	8.5000	15.0000	15.0000	cluster_0	H03	H0316	H031608	H03160817
P0007	2.0000	22.0000	9.5000	cluster_4	H03	H0313	H031305	H03130519
P0008	5.0000	16.0000	5.0000	cluster_0	H00	H0000	H000004	H00000400

Fuente d... Hoja 1 Null Scanner Sales

Module 1: Preparing Data and Creating Basic Visualizations

Clean the Product Hierarchy Data . Check for missing values

¿Necesita ayuda para añadir datos? [Más información](#)

product_hierarchy.csv

10 campos 699 filas

100

→

filas

⚙️

▼

Nombre

product_hierarchy.csv

Campos

Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	product_id	product_hi...	product_id
#	product_length	product_hi...	product_length
#	product_depth	product_hi...	product_depth
#	product_width	product_hi...	product_width

Abc	#	#	#	Abc	Abc	Abc	Abc	Abc
product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv
product_id	product_length	product_depth	product_width	cluster_id	hierarchy1_id	hierarchy2_id	hierarchy3_id	hierarchy4_id
P0000	5.0000	20.0000	12.0000	NULL	H00	H0004	H000401	H00040105
P0001	13.5000	22.0000	20.0000	cluster_5	H01	H0105	H010501	H01050100
P0002	22.0000	40.0000	22.0000	cluster_0	H03	H0315	H031508	H03150800
P0004	2.0000	13.0000	4.0000	cluster_3	H03	H0314	H031405	H03140500
P0005	16.0000	30.0000	16.0000	cluster_9	H03	H0312	H031211	H03121109
P0006	8.5000	15.0000	15.0000	cluster_0	H03	H0316	H031608	H03160817
P0007	2.0000	22.0000	9.5000	cluster_4	H03	H0313	H031305	H03130519
P0008	5.0000	16.0000	5.0000	cluster_0	H00	H0000	H000004	H00000400

¿Necesita ayuda para añadir datos? [Más información](#)

product_hierarchy.csv

17 campos 699 filas

100

→

filas

⚙️

▼

Nombre

product_hierarchy.csv

Campos

Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	product_id	product_hi...	product_id
Abc	ProductId_New	Cálculo	Calculation_151...
#	product_length	product_hi...	product_length
#	product_depth	product_hi...	product_depth

product_hierarchy.csv	Abc	Abc	Abc	Abc	#	Abc	Abc	Abc
product_hierarchy.csv	Cálculo	product_hierarchy.csv	Cálculo	product_hierarchy.csv	Cálculo	product_hierarchy.csv	Cálculo	product_hierarchy.csv
product_id	Hierarchy1_New	hierarchy2_id	Hierarchy2_New	hierarchy3_id	Hierarchy3_New	hierarchy4_id	Hierarchy4_New	hierarchy5_id
H00	H0004	H0004	H0004	H000401	NULL	H00040105	H00040105	H0004010534
H01	H0105	H0105	H0105	H010501	NULL	H01050100	H01050100	H0105010006
H03	H0315	H0315	H0315	H031508	NULL	H03150800	H03150800	H0315080028
H03	H0314	H0314	H0314	H031405	NULL	H03140500	H03140500	H0314050003
H03	H0312	H0312	H0312	H031211	NULL	H03121109	H03121109	H0312110917
H03	H0316	H0316	H0316	H031608	NULL	H03160817	H03160817	H0316081708
H03	H0313	H0313	H0313	H031305	NULL	H03130519	H03130519	H0313051904
H00	H0000	H0000	H0000	H000004	NULL	H00000400	H00000400	H0000040017

Module 1: Preparing Data and Creating Basic Visualizations

Clean the Product Hierarchy Data . Format the data

product_hierarchy.csv

17 campos 699 filas

100

→

filas

Nombre

product_hierarchy.csv

Campos

Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	product_id	product_hi...	product_id
Abc	ProductId_New	Cálculo	Calculation_151...
#	product_length	product_hi...	product_length
#	product_depth		

Abc	Abc	#	#	#	Abc	Abc	Abc	Abc
product_hierarchy.	Cálculo	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	Cálculo	product_hierarchy.csv	Cálculo
product_id	ProductId_New	product_length	product_depth	product_width	cluster_id	ClusterId_New	hierarchy1_id	Hierarchy1_
P0000	P0000	5.0000	20.0000	12.0000	NULL	NOT_CLUSTER	H00	H00
P0001	P0001	13.5000	22.0000	20.0000	cluster_5	CLUSTER_5	H01	H01
P0002	P0002	22.0000	40.0000	22.0000	cluster_0	CLUSTER_0	H03	H03
P0004	P0004	2.0000	13.0000	4.0000	cluster_3	CLUSTER_3	H03	H03
P0005	P0005	16.0000	30.0000	16.0000	cluster_9	CLUSTER_9	H03	H03
P0006	P0006	8.5000	15.0000	15.0000	cluster_0	CLUSTER_0	H03	H03

product_hierarchy.csv

17 campos 699 filas

100

→

fi

Nombre

product_hierarchy.csv

Campos

Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	product_id	product_hi...	product_id
Abc	ProductId_New	Cálculo	Calculation_151...
#	product_length	product_hi...	product_length
#	product_depth	product_hi...	product_depth

Abc	Abc	#	#	#	Abc	Abc	Abc	Abc
product_hierarchy.	Cálculo	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	product_hierarchy.csv	Cálculo	product_hierarchy.csv	product_hierarchy.csv
product_id	ProductId_New	product_length	product_depth	product_width	cluster_id	ClusterId_New	hierarchy1_id	
P0000	P0000	5.0000	20.0000	12.0000	NULL	NOT_CLUSTER	H00	
P0001	P0001	13.5000	22.0000	20.0000	cluster_5	CLUSTER_5	H01	
P0002	P0002	22.0000	40.0000	22.0000	cluster_0	CLUSTER_0	H03	
P0004	P0004	2.0000	13.0000	4.0000	cluster_3	CLUSTER_3	H03	
P0005	P0005	16.0000	30.0000	16.0000	cluster_9	CLUSTER_9	H03	
P0006	P0006	8.5000	15.0000	15.0000	cluster_0	CLUSTER_0	H03	
P0007	P0007	2.0000	22.0000	9.5000	cluster_4	CLUSTER_4	H03	
P0008	P0008	5.0000	16.0000	5.0000	cluster_0	CLUSTER_0	H00	

Clean the Product Hierarchy Data . Standardize text

product_hierarchy.csv

17 campos 699 filas

100 → filas

Nombre

product_hierarchy.csv

Campos

Tipo	Nombre de campo	Tabla física	Nombre de ca...
Abc	product_id	product_hi...	product_id
Abc	ProductId_New	Cálculo	Calculation_151...
#	product_length	product_hi...	product_length
#	product_depth	product_hi...	product_depth

product_hierarchy.csv	Abc Cálculo	Abc product_hierarchy.csv	Abc Cálculo	Abc product_hierarchy.csv	# Cálculo	Abc product_hierarchy.csv	Abc Cálculo	Abc product_hierarchy.csv
product_id	Hierarchy1_New	product_hierarchy2_id	Hierarchy2_New	product_hierarchy3_id	Hierarchy3_New	product_hierarchy4_id	Hierarchy4_New	product_hierarchy5_id
H00	H0004	H0004	H0004	H000401	NULL	H00040105	H00040105	H0004010534
H01	H0105	H0105	H0105	H010501	NULL	H01050100	H01050100	H0105010006
H03	H0315	H0315	H0315	H031508	NULL	H03150800	H03150800	H0315080028
H03	H0314	H0314	H0314	H031405	NULL	H03140500	H03140500	H0314050003
H03	H0312	H0312	H0312	H031211	NULL	H03121109	H03121109	H0312110917
H03	H0316	H0316	H0316	H031608	NULL	H03160817	H03160817	H0316081708
H03	H0313	H0313	H0313	H031305	NULL	H03130519	H03130519	H0313051904
H00	H0000	H0000	H0000	H000004	NULL	H00000400	H00000400	H0000040017

Module 1: Preparing Data and Creating Basic Visualizations

Cleaning All Files `allotherCSVfiles_imported`

Tableau Public - Libro1

Archivo Datos Ventana Ayuda

Tableau Desktop Public Edition

Conexiones

Añadir

product_names

Archivos

☐ Usar el intérprete de datos

Puede que el intérprete de datos esté disponible para limpiar su libro de trabajo

Archivo de texto.

city_names.csv

product_hierarchy.csv

product_names.csv

sales.csv

store_cities.csv

store_names.csv

product_names

product_names.csv

Relacionar tablas

Arrastre tablas cerca product_names.csv para establecer relaciones entre ellas

Añadir una tabla base

Arrastre tablas base adicionales para el análisis multifactor.

¿Necesita ayuda para añadir datos? [Más información](#)

Module 1: Preparing Data and Creating Basic Visualizations

sales

Create the relationships
Before and after

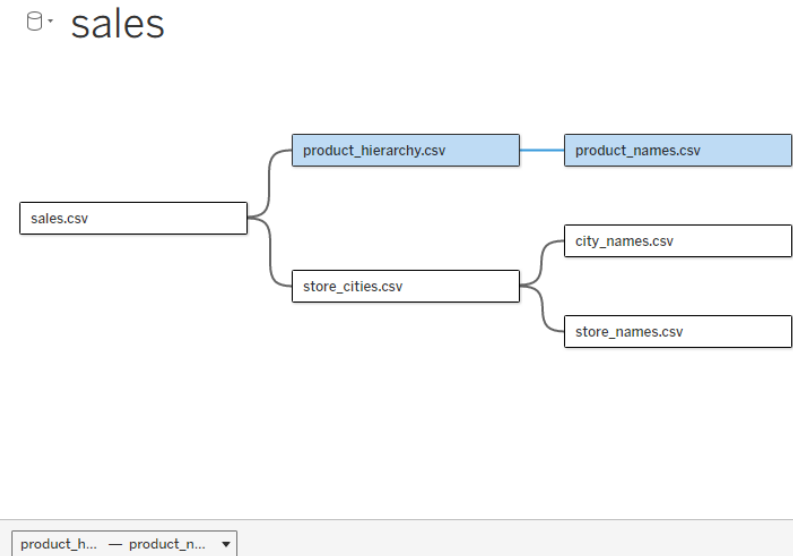
sales.csv



Module 1: Preparing Data and Creating Basic Visualizations

Data Interpreter

The screenshot shows the Data Interpreter interface. On the left, a sidebar contains two sections: 'Conexiones' (Connections) with a list of 'sales' and 'product_hierarchy', and 'Archivos' (Files) with a list of 'city_names.csv', 'product_hierarchy.csv', 'product_names.csv', 'sales.csv', 'store_cities.csv', and 'store_names.csv'. The main area displays a hierarchical diagram of the 'sales' dataset. The diagram shows 'sales.csv' branching into 'product_hierarchy.csv' and 'store_cities.csv'. 'product_hierarchy.csv' further branches into 'product_names.csv'. 'store_cities.csv' branches into 'city_names.csv' and 'store_names.csv'. A dropdown menu at the bottom of the sidebar shows 'product_h...' and 'product_n...'. A modal window is open over the diagram, showing the same sidebar content and a 'Limpiado con el intérprete de datos' (Cleaned with the data interpreter) checkbox, which is checked. Below the checkbox, there is a link 'Revise los resultados' (Review the results) and a note about undoing changes.



Module 1: Preparing Data and Creating Basic Visualizations

Review the results

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1																				
2																				
3																				
4																				
5																				
6																				
7																				
8																				
9																				
10																				
11																				
12																				
13																				
14																				
15																				
16																				
17																				

<

>

Clave para el intérprete de dat

product_hierarchy_csv

product_names_csv

store_c ...

+

:

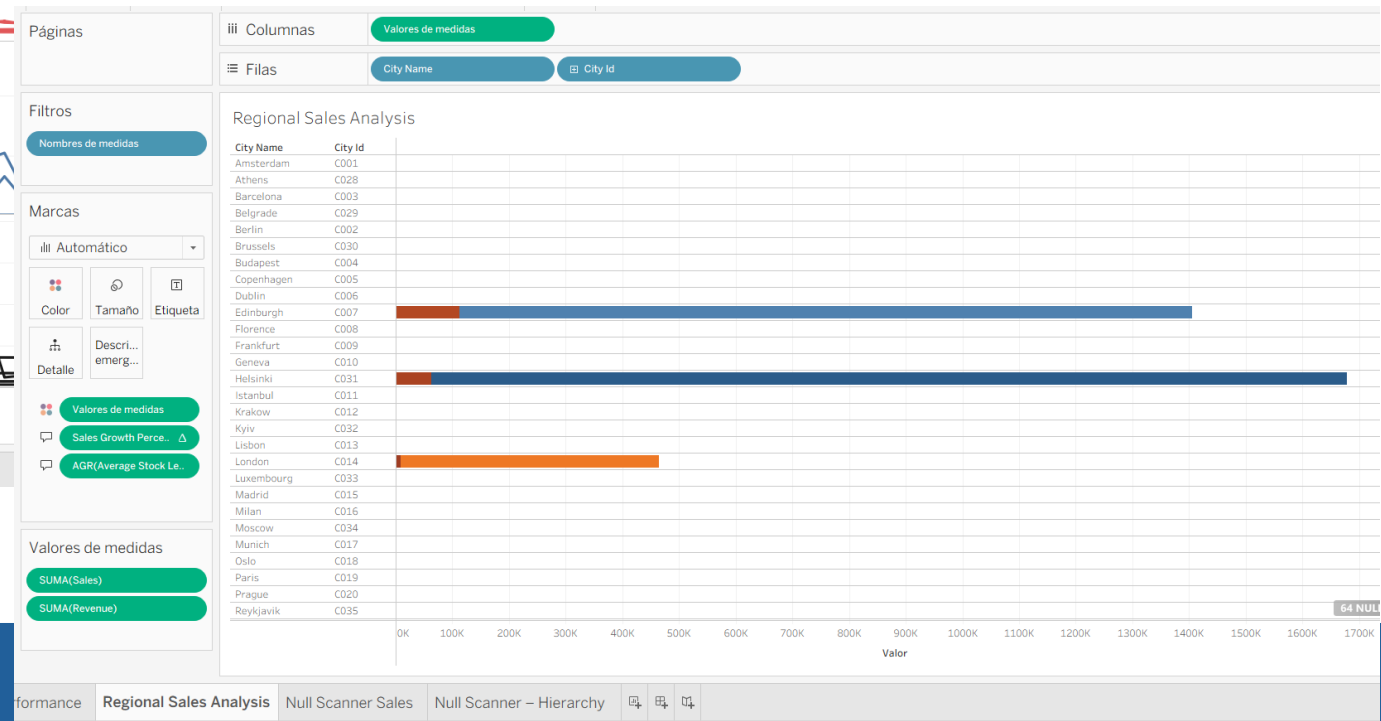
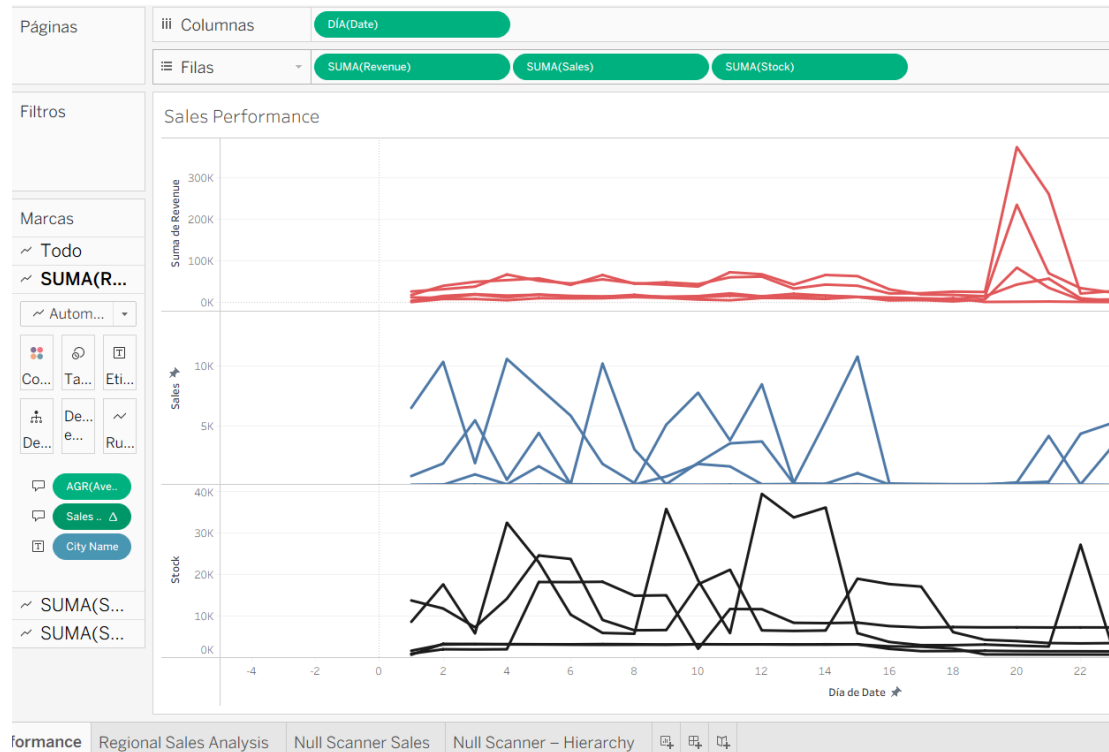
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▶

Module 2: Advanced Visualizations, Dashboards, and Stories with Tableau

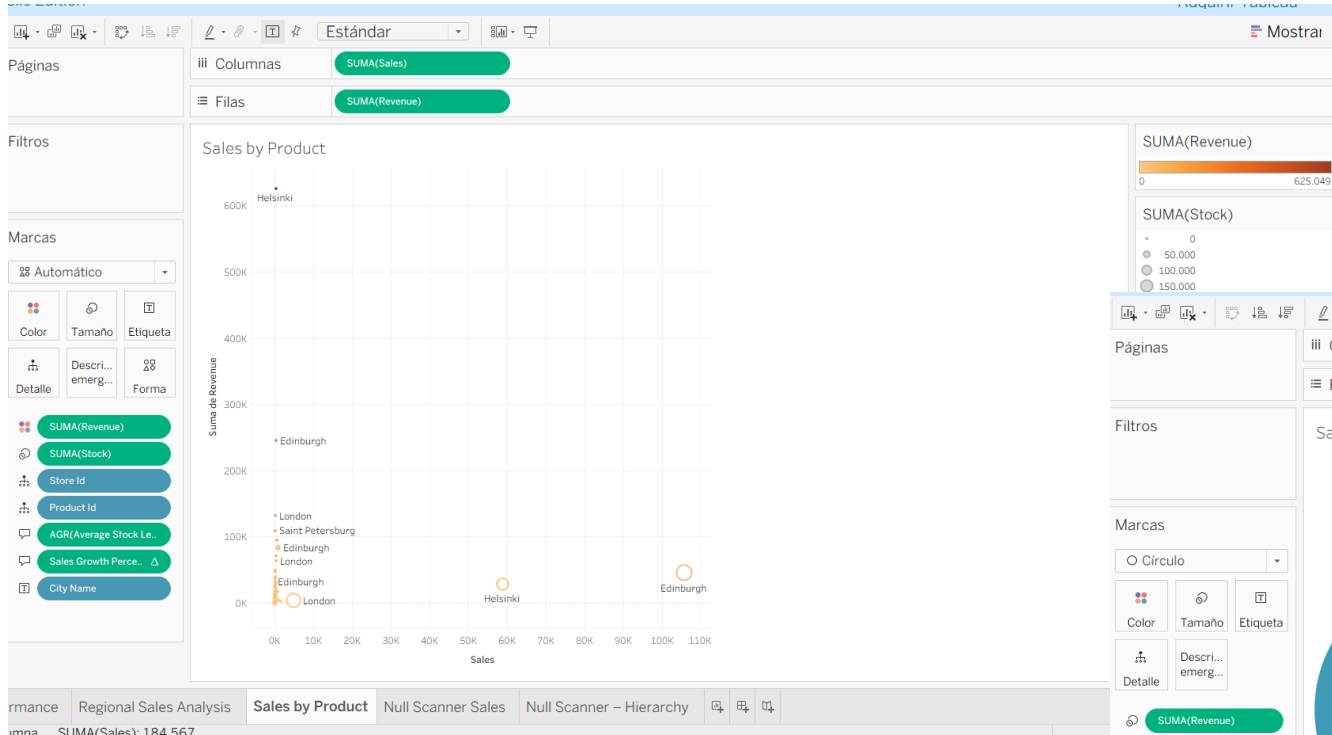
Basic Tableau Visualizations

- Sales Performance (line chart, Day):** Position along a common time axis makes **trends and seasonality** instantly visible; separate panes for Sales, Revenue, and Stock avoid scale confusion and spotlight **spikes, dips, and stockouts**.
- Regional Sales Analysis (horizontal bars):** Bar **length** enables fast **city-to-city comparison**; sorted bars surface **top/bottom performers** at a glance.

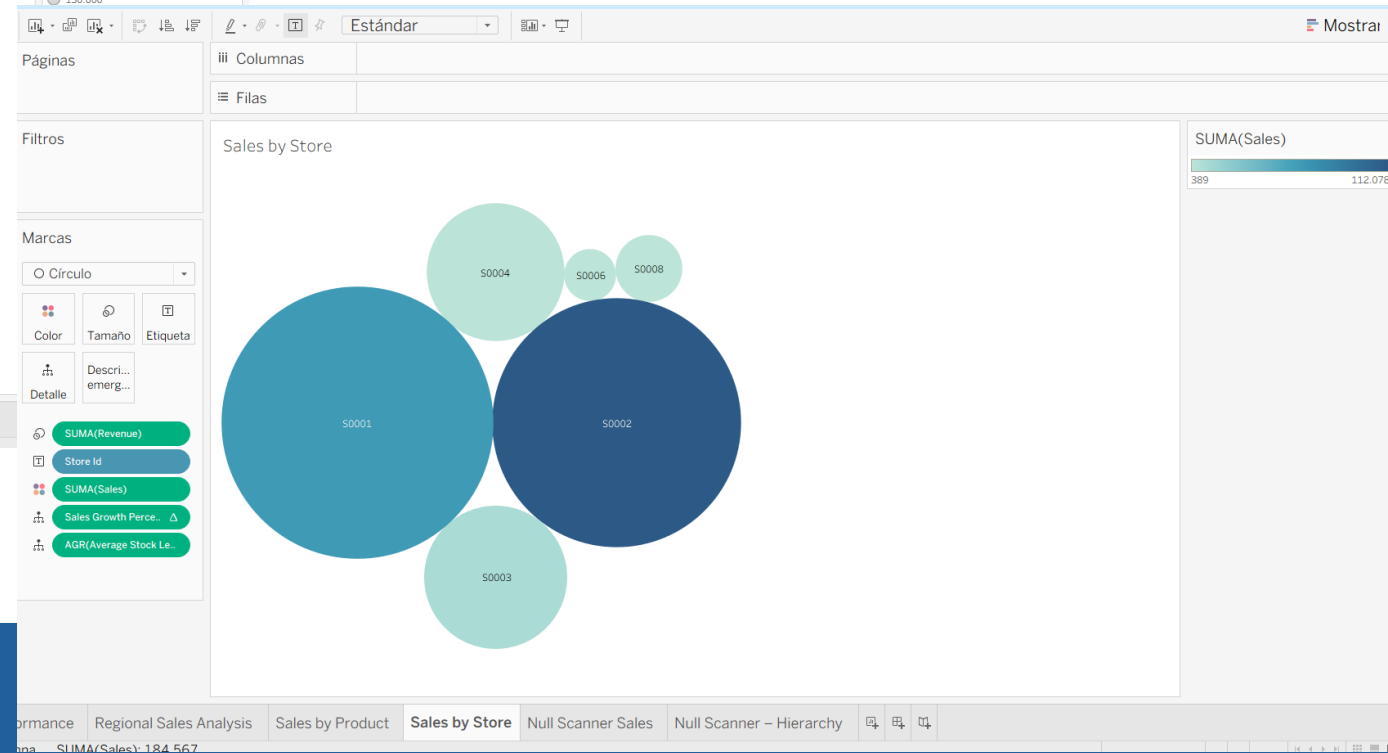


Module 2: Advanced Visualizations, Dashboards, and Stories with Tableau

Basic Tableau Visualizations

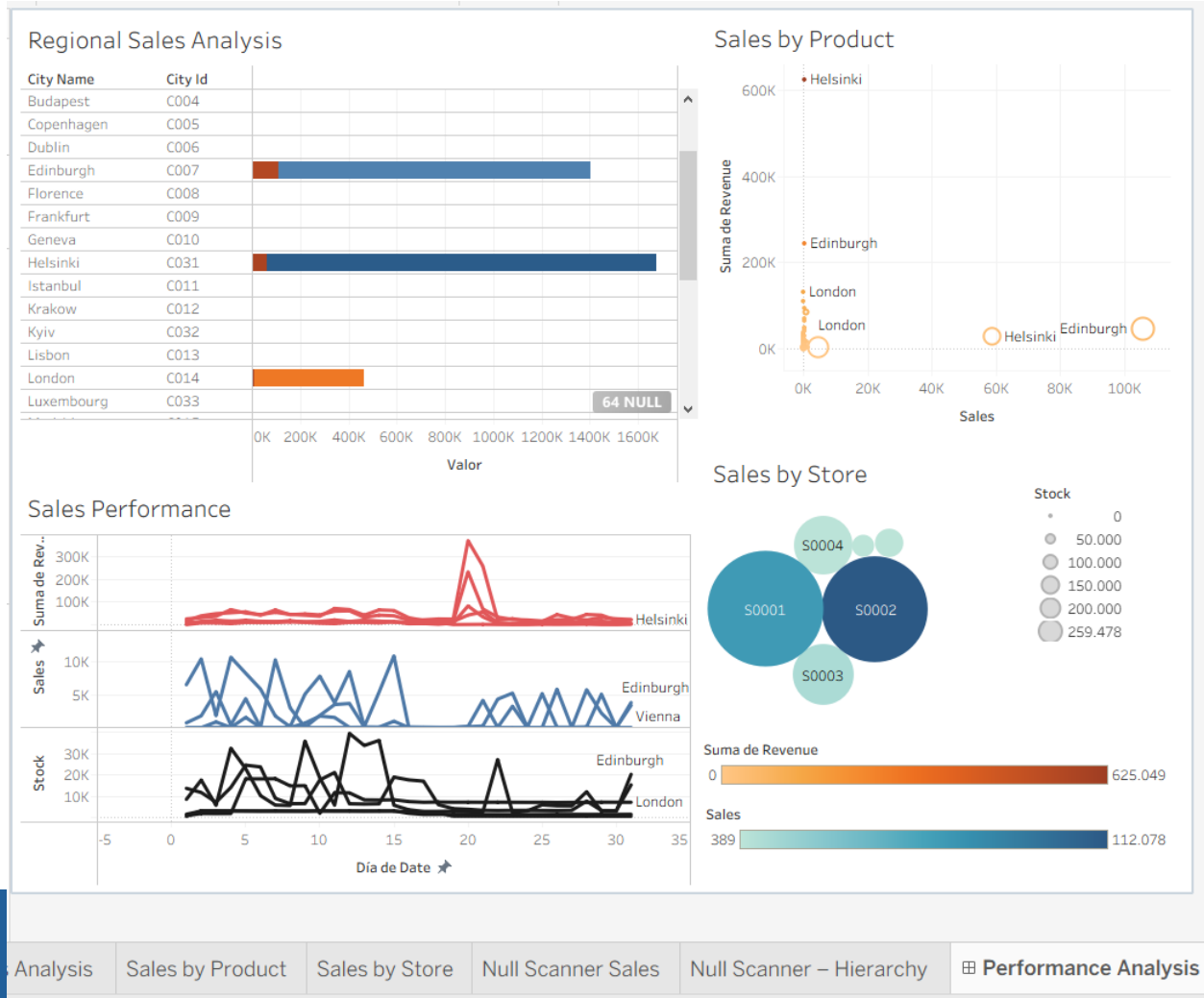


- **Sales by Product (scatter): Position (Sales vs. Revenue)** reveals conversion efficiency; **Color (Revenue)** and **Size (Stock)** add two extra layers for **high-value/overstock outliers**; labels give immediate context.
- **City_Hierarchy drill-down:** From **City** → **Store** → **Product** keeps the story consistent across levels, supporting both **overview and detail** without changing the canvas.
- **Aligned tooltips with KPIs:** Adding *Sales Growth %* and *Average Stock* to tooltips turns every mark into a **mini-report**, reducing the need for separate tables.



Module 2: Advanced Visualizations, Dashboards, and Stories with Tableau

Advanced Visualizations Using Tableau



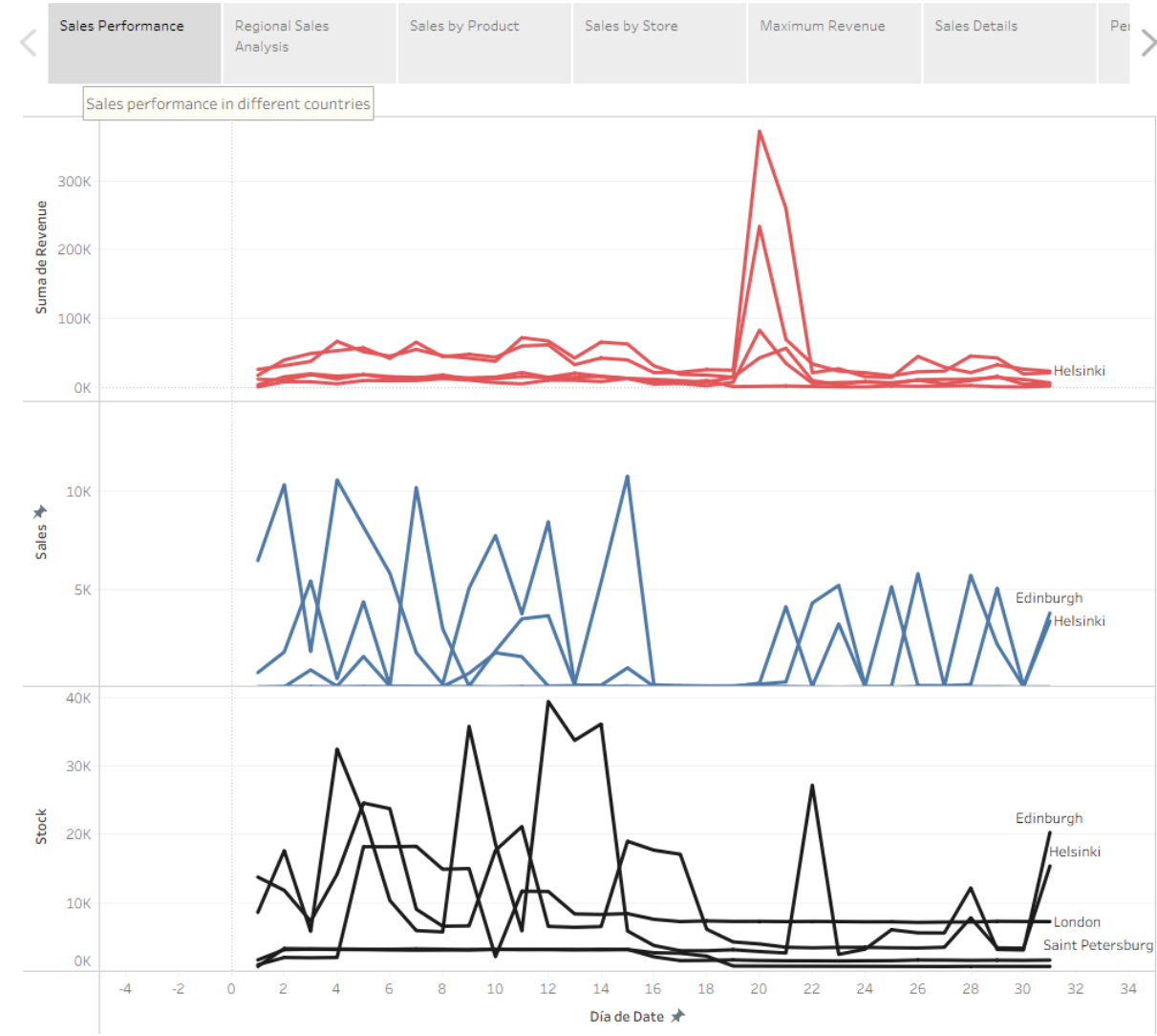
The visualizations speed up decision-making by showing trends (lines), top/bottom cities (bars), product outliers (scatter), and stock risks (bubbles). This enables prioritizing promotions, reallocating inventory, and focusing on the highest-impact stores.



Module 3: Tableau Capstone Project Submission and Evaluation

- 1st Story point

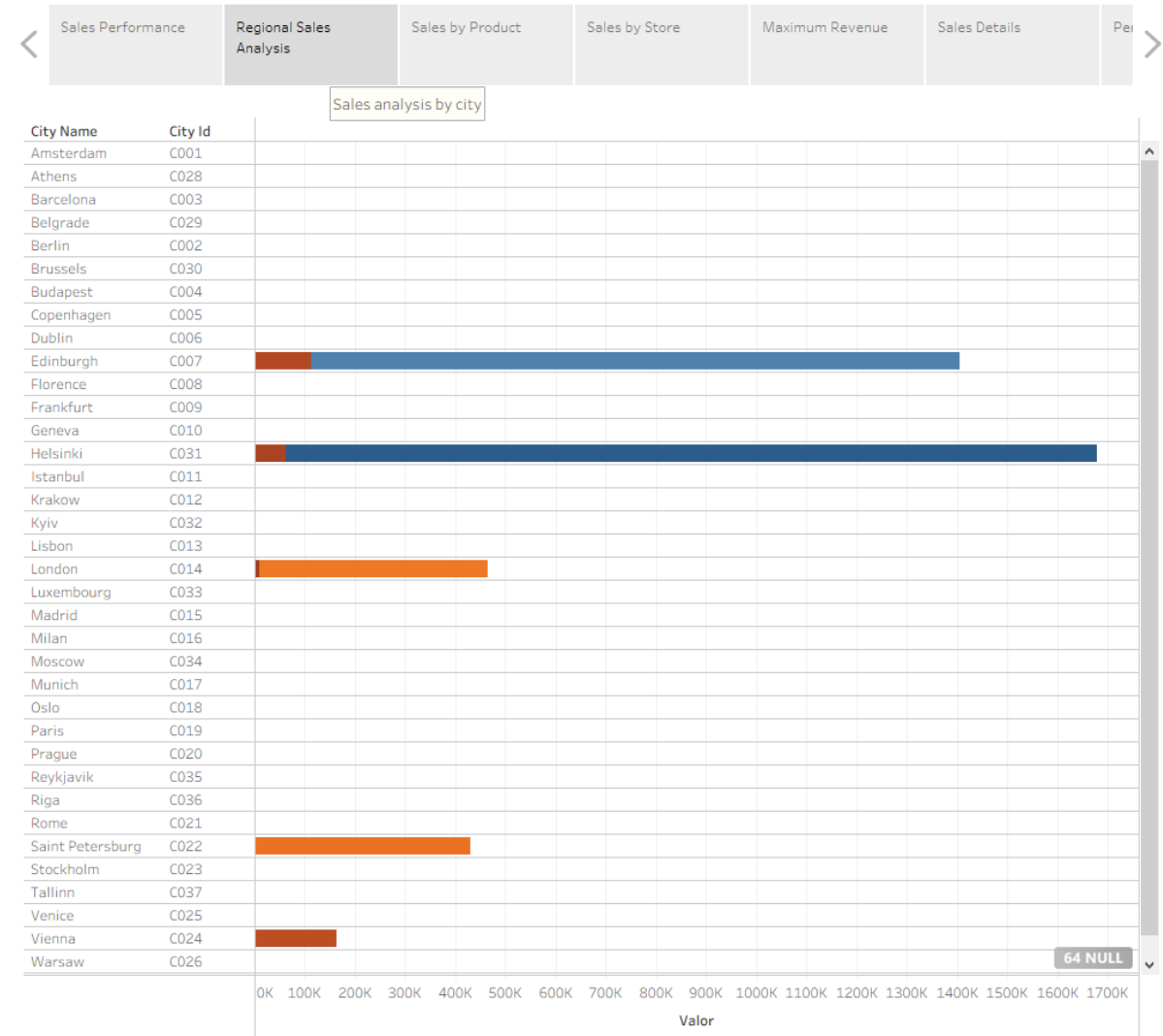
Tableau Capstone Story



Module 3: Tableau Capstone Project Submission and Evaluation

- 2nd Story point

Tableau Capstone Story



Module 3: Tableau Capstone Project Submission and Evaluation

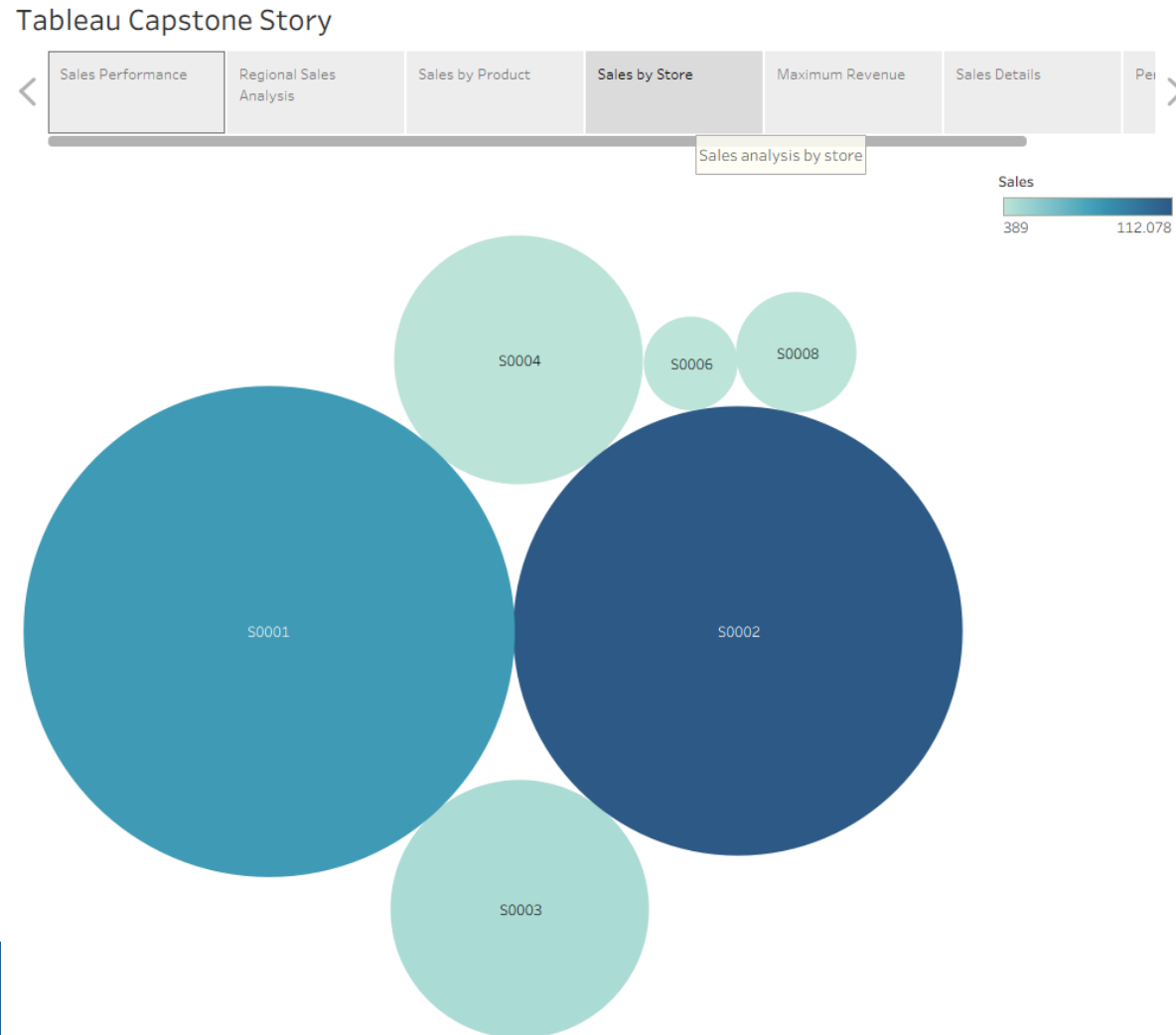
- 3rd Story point

Tableau Capstone Story



Module 3: Tableau Capstone Project Submission and Evaluation

- 4th Story point



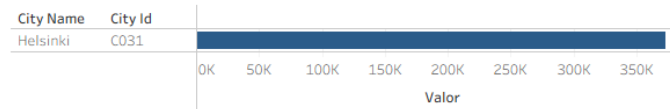
Module 3: Tableau Capstone Project Submission and Evaluation

- 5th Story point

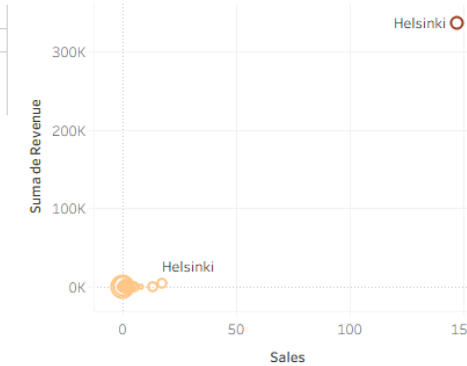
Tableau Capstone Story



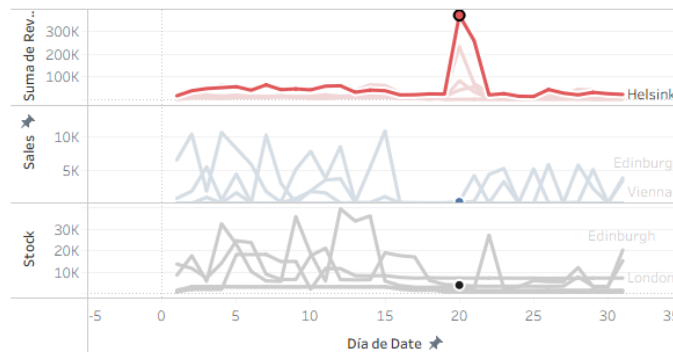
Regional Sales Analysis



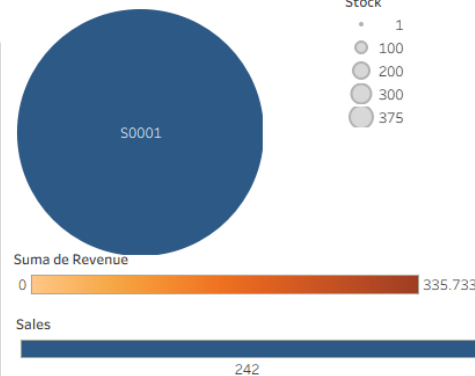
Sales by Product



Sales Performance



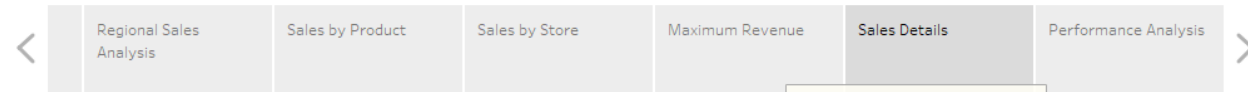
Sales by Store



Module 3: Tableau Capstone Project Submission and Evaluation

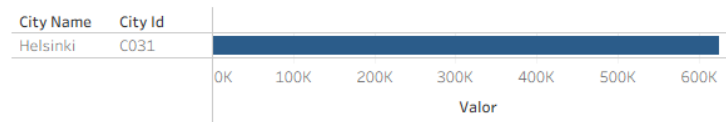
- 6th Story point

Tableau Capstone Story

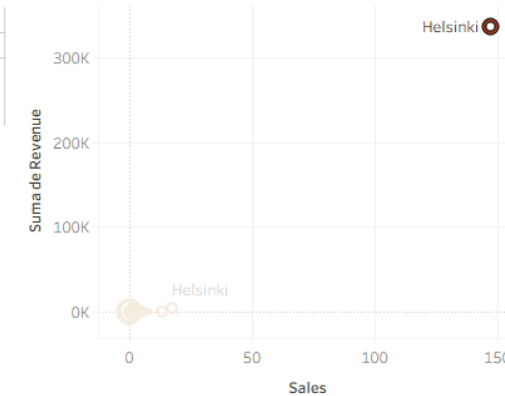


Sales Detail of store Mediagalaxy

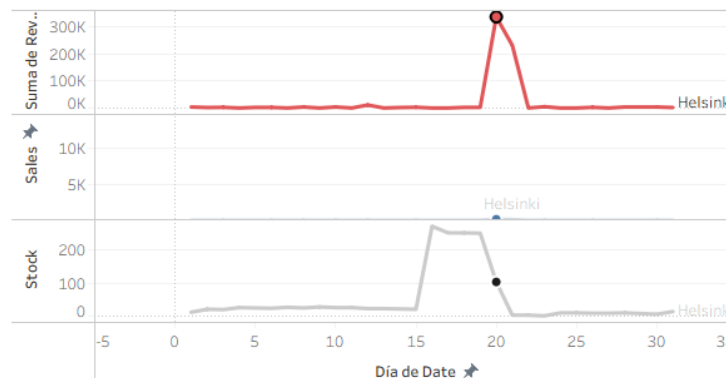
Regional Sales Analysis



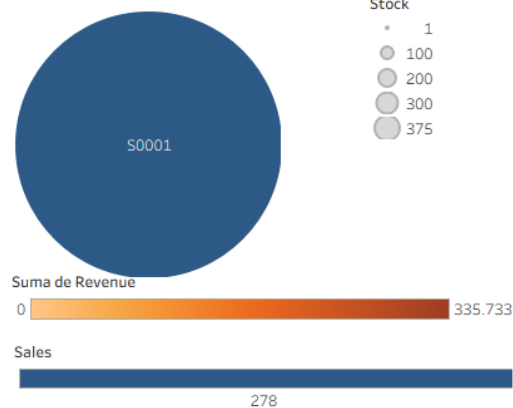
Sales by Product



Sales Performance



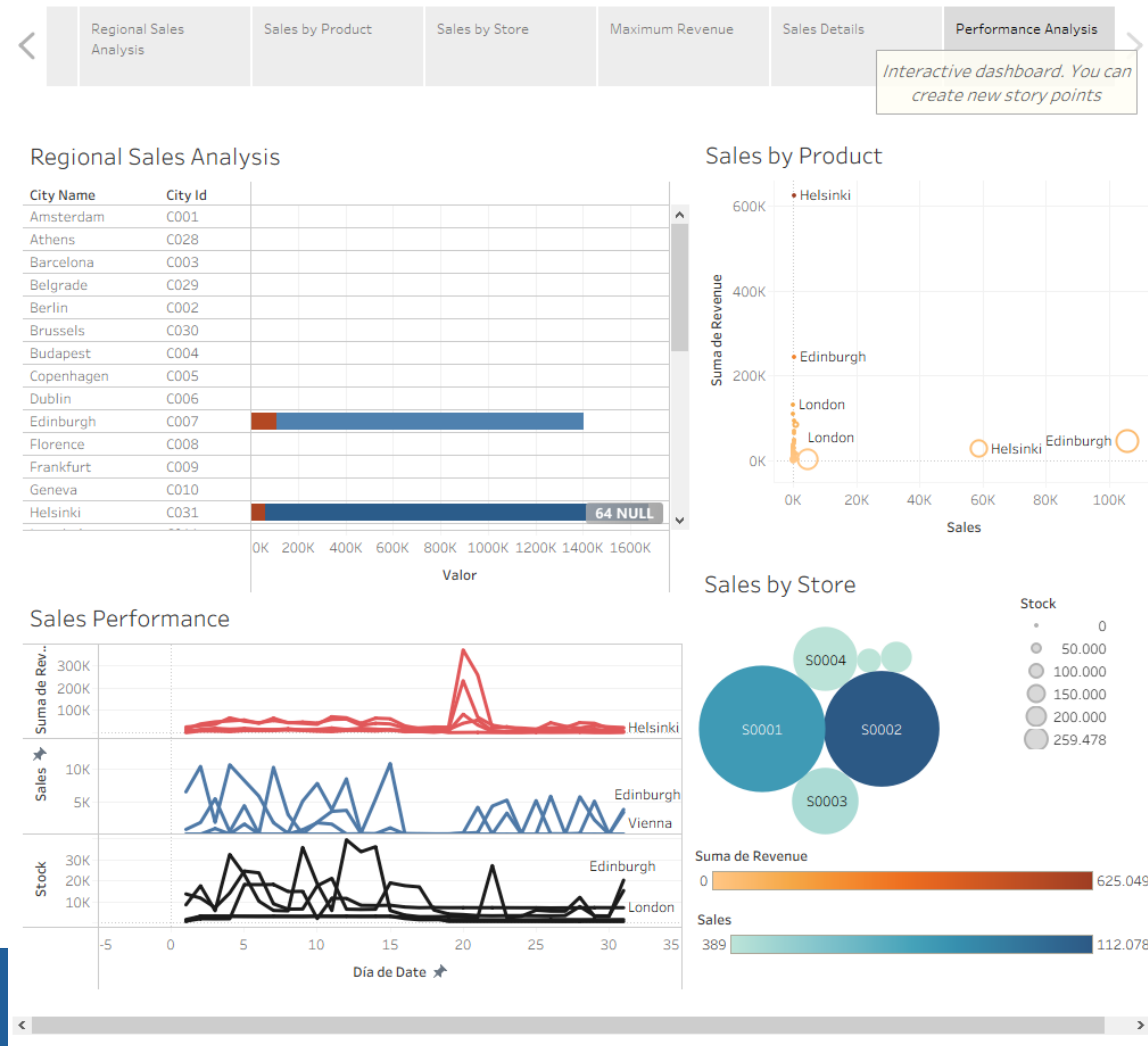
Sales by Store



Module 3: Tableau Capstone Project Submission and Evaluation

- 7th Story point

Tableau Capstone Story



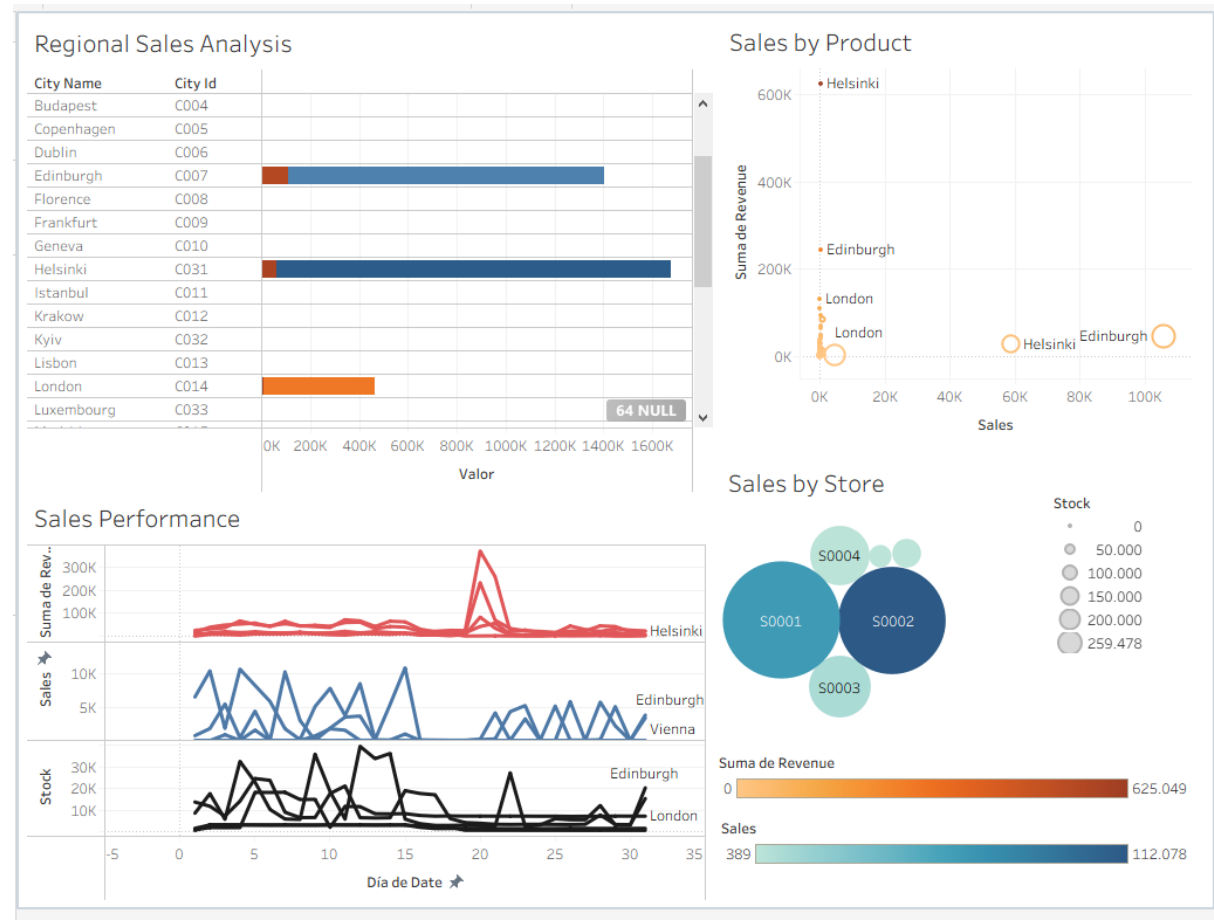


Results

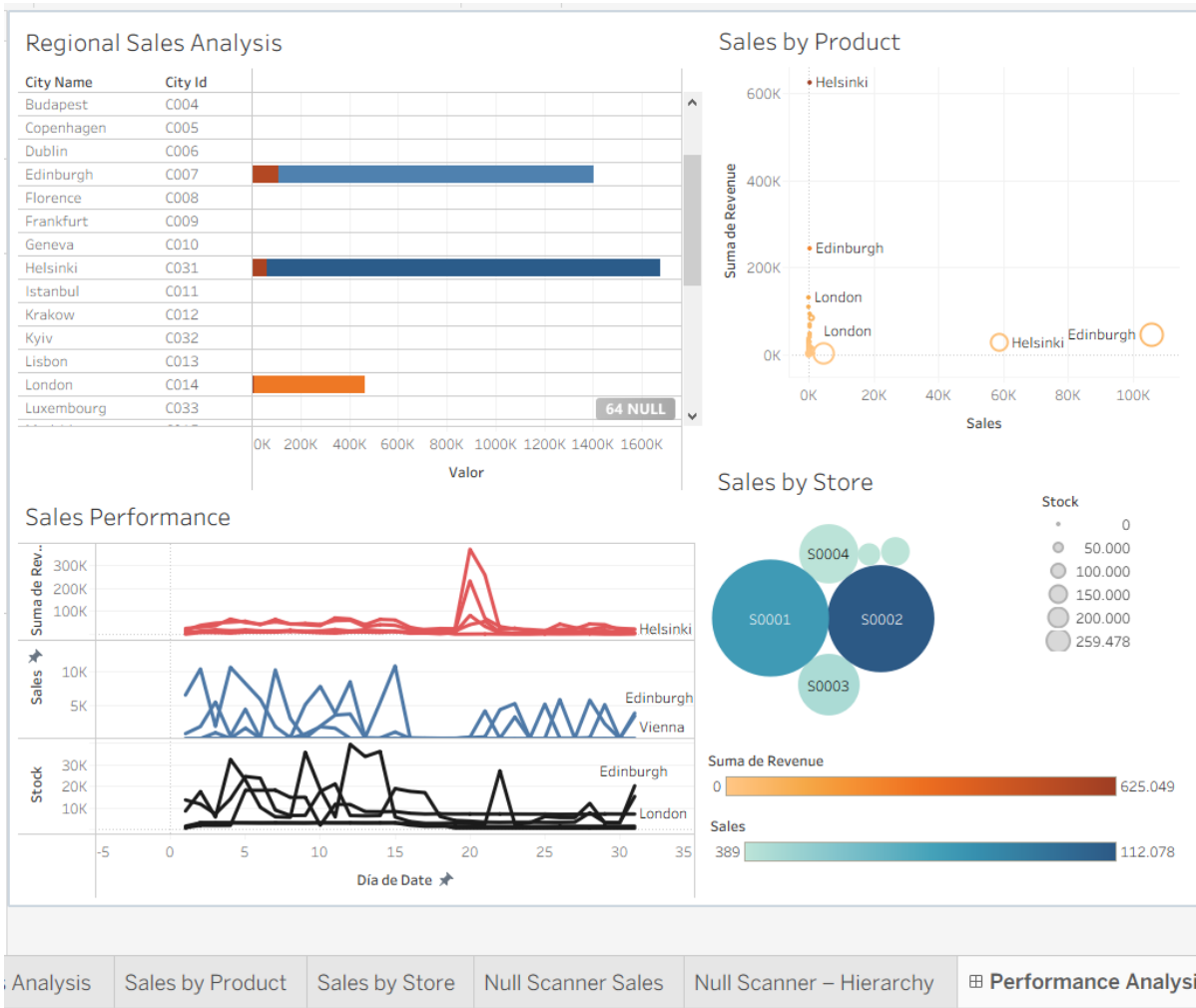
Module 1, Lesson 2: Importing and Cleaning Data

- **Screenshots have already been added on the previous slides.**
 - the screenshots taken from Module 1 – Lesson 2 after data cleaning of sales.csv file
 - the screenshots taken from Module 1 – Lesson 2 after data cleaning of product_hierarchy.csv
 - the screenshot taken from Module 1 – Lesson 2 after cleaning all files using Data Interpreter
 - the screenshots taken from Module 2 – Lesson 1 after creating worksheets using Tableau
 - the screenshots taken from Module 2 – Lesson 2 after creating a dashboard
 - the screenshots taken from Module 3 – Lesson 1 after creating a Story and giving it a title

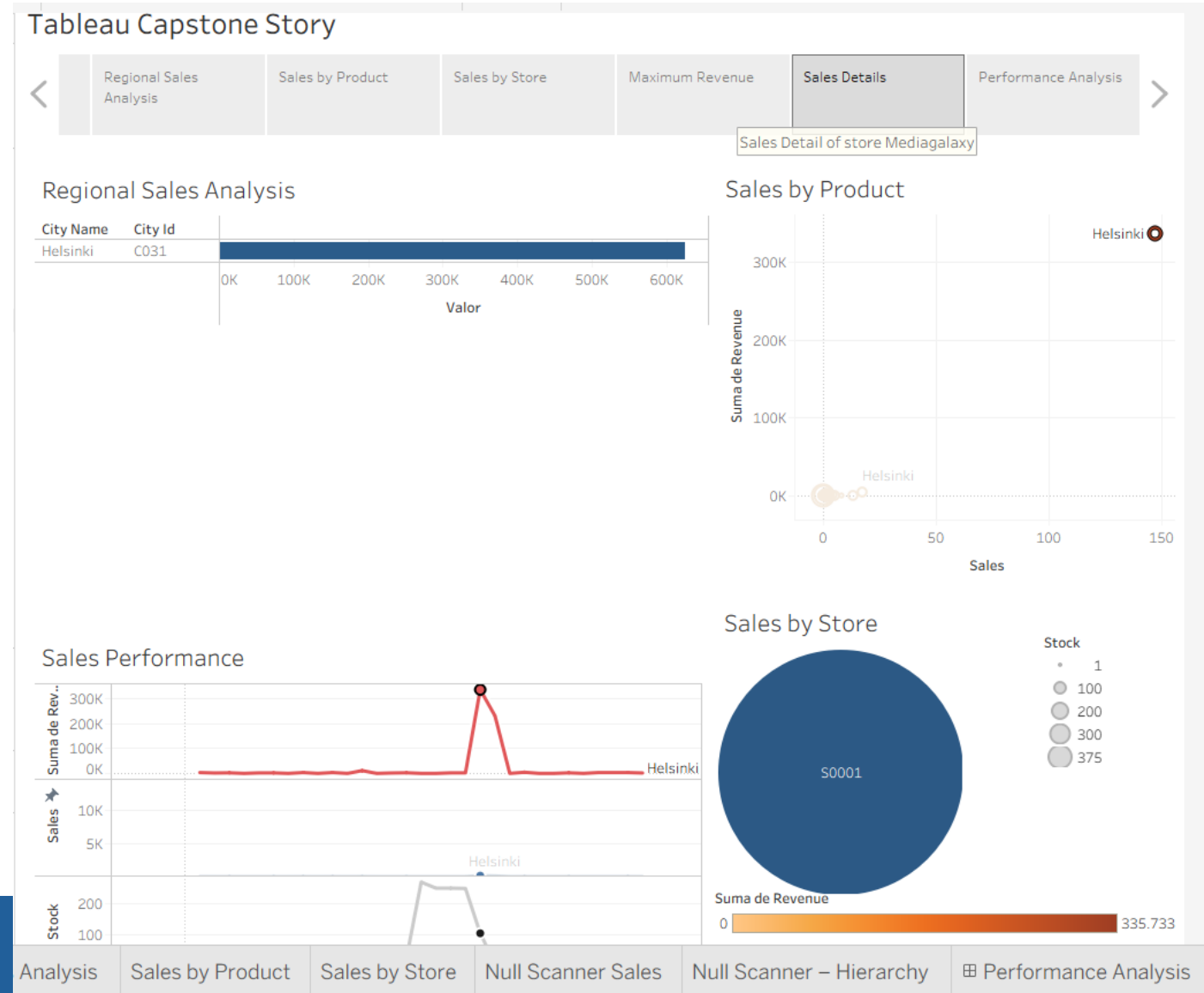
Module 1, Lesson 3: Basic Tableau Visualizations



Module 2, Lesson 1: Advanced Visualizations Using Tableau



Module 2, Lesson 2: Creating Engaging Stories in Tableau





Extra materials

Presentation Improvement

Innovative Presentation Ideas:

- What-if controls (parameters) to simulate discounts/promotions and see instant impact on sales and inventory.
- Role-based views using dynamic zone visibility: “Executive” (KPIs/trends) vs. “Ops” (stock/stockouts).
- Visual alerts: reference lines/thresholds and traffic-light color for Growth % and Stock.
- Quick anomaly detection: Z-score highlight table or confidence bands on the time series.
- Story Points: 5–6 steps to narrate (Data Quality → Trends → Top/Bottom Cities → Product Outliers → Recommendations).

Innovation and Value Addition

Value Added:

- Reliable, reusable model: normalized IDs, unified types/formats, logical relationships without duplication, and full before/after audit trail.
- Actionable KPIs: embedded Sales Growth % and Average Stock in tooltips; cross-checks across sales—revenue—stock and null scanners.
- Integrated storytelling: a single dashboard with filter actions and City_Hierarchy (City→Store→Product) for seamless zoom-in from overview to detail.



Conclusion

Summary

The dashboard surfaces daily trends, top/bottom cities & stores, and product outliers (high sales with low revenue or overstock).

Filter actions let you validate hypotheses in seconds (promo effects, seasonality, stockouts) and prioritize decisions: targeted promos, inventory rebalancing, and focus on high-impact stores.

Next steps: extend the time window, add cost/margin, build forecasting, and automate threshold-based alerts



Appendix

Appendix

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